

Climate Change

Causes, impacts, solutions and real examples

Introduction

Climate change means long-term changes in Earth's temperature, rainfall, wind patterns and seasons. The main reason for climate change today is human activity. Burning coal, petrol, diesel, oil and natural gas for electricity, transport and factories releases greenhouse gases such as carbon dioxide.

These gases trap heat in the atmosphere. A small greenhouse effect keeps Earth warm enough for life, but extra gases trap too much heat and make the planet warmer. As a result, weather patterns are becoming more extreme and less predictable.

How climate change impacts Earth and the environment

Rising temperature leads to stronger and longer heatwaves. Glaciers and polar ice melt, contributing to sea-level rise. Coastal areas can then face erosion, flooding and damage to homes, roads and other infrastructure.

Rainfall patterns are also changing. Some places receive very heavy rain in a short time, causing floods and landslides. Other places receive too little rain and face drought. These changes affect drinking water, farming, food production and electricity generation.

Forests, oceans, animals and plants are affected too. Hotter oceans can damage coral reefs, while changes in temperature and rainfall can force species to move from their natural habitats. Farmers may struggle to grow crops when seasons are irregular.

Problems caused by climate change

- Heatwaves that can lead to dehydration, illness and loss of work or study time.
- Heavy rain, floods and landslides that can damage homes, roads, schools and hospitals.
- Droughts that reduce water availability and harm agriculture.
- Sea-level rise that threatens coastal cities and villages.
- Crop loss, higher food prices and greater risk for poor and vulnerable communities.

Real Examples and Solutions

Real examples

India experienced several extreme weather events in 2025. Heatwave conditions were reported in parts of Rajasthan, Gujarat, central India and northern India before the monsoon. Heavy rainfall, floods, landslides and cloudbursts were also reported in different regions. This shows that one country can face both extreme heat and intense rainfall in the same year.

Cyclone Montha crossed the Andhra Pradesh coast in October 2025. It brought extremely heavy rain to Coastal Andhra Pradesh and heavy to very heavy rain in nearby regions, including Telangana and Odisha. The event showed why early warnings, evacuation plans and safe shelters are essential.

Europe also faced repeated heatwaves during the summer of 2025. Many areas experienced severe heat stress, drought conditions, wildfire risk and pressure on water supplies.

Climate change and daily life

Climate change affects normal daily activities. During heatwaves, people may find it difficult to travel, study or work outdoors. Heavy rainfall can interrupt transport, damage roads and close schools or colleges. Farmers are especially affected when rain comes too early, too late or very heavily in a short time.

Students can help by saving electricity and water, reducing waste, carrying reusable bottles and bags, and supporting environmental activities in their college. At the same time, governments and companies must take larger action because major changes in energy, transport and industry are needed.

How to solve the problem

Who can act	Examples of action
Governments	Invest in solar and wind energy, improve public transport, protect forests and create strong warning systems.
Companies	Use cleaner technology, save energy, reduce industrial pollution and cut waste.
Colleges	Save electricity, reduce plastic, manage waste properly and build environmental awareness.
Individuals	Save water and energy, use public transport where possible, avoid waste and use reusable products.

Conclusion

Climate change is already affecting people and the environment. Rising temperatures, floods, droughts, heatwaves and cyclones show that the issue is real and urgent. However, it can be managed through cleaner energy, better planning, protection of nature, early warnings and cooperation between governments, businesses, colleges and citizens.

The main message is simple: climate change is a serious problem, but informed action today can reduce damage and create a safer future.